

Ubuntu DHCP 서버 구축 실습 정리

1. DHCP 패키지 설치

apt install -y isc-dhcp-server 명령어를 사용하여 isc-dhcp-server 패키지를 설치

```
To restore this content, you can run the 'unminimize' command.
Last login: Tue Sep  9 08:49:02 2025 from 172.16.23.1
root@sheep:~# apt install -y isc-dhcp-server
```

2. DHCP 설정 파일 수정

설정 파일 (/etc/dhcp/dhcpd.conf)을 편집기(nano or vim)에서 열어 사진과 같이 net ID 와 서브넷, 할당할 ip 범위, 게이트웨이, DNS 서버를 차례로 지정

```
# fixed-address fantasia.example.com;
#}

# You can declare a class of clients and then do address allocation
# based on that.  The example below shows a case where all clients
# in a certain class get addresses on the 10.17.224/24 subnet, and all
# other clients get addresses on the 10.0.29/24 subnet.

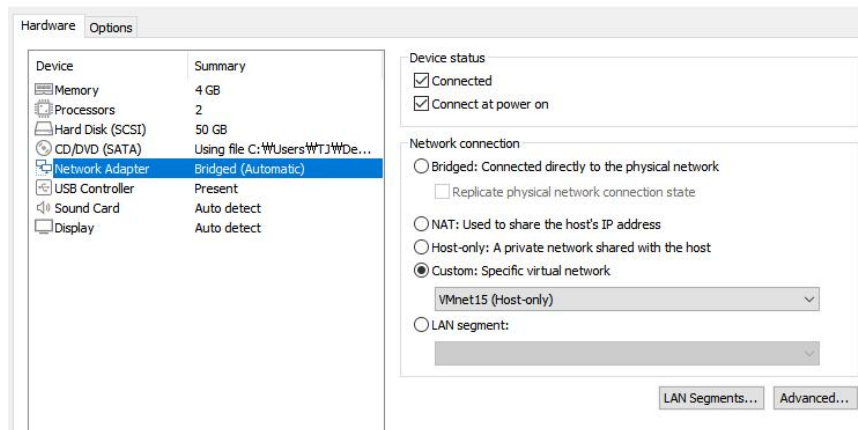
#class "foo" {
#  match if substring (option vendor-class-identifier, 0, 4) = "SUNW";
#}

#shared-network 224-29 {
#  subnet 10.17.224.0 netmask 255.255.255.0 {
#    option routers rtr-224.example.org;
#  }
#  subnet 10.0.29.0 netmask 255.255.255.0 {
#    option routers rtr-29.example.org;
#  }
#  pool {
#    allow members of "foo";
#    range 10.17.224.10 10.17.224.250;
#  }
#  pool {
#    deny members of "foo";
#    range 10.0.29.10 10.0.29.230;
#  }
#}
subnet 172.16.23.0 netmask 255.255.255.0 {
  range 172.16.23.20 172.16.23.50;
  option routers 172.16.23.30;
  option domain-name-servers 8.8.8.8;
}

"/etc/dhcp/dhcpd.conf" 118L, 3815B
```

3. 네트워크 설정 변경 (VMware)

네트워크 어댑터 모드를 Bridged 에서 Custom (VMnet15, Host-only)로 변경 후



DHCP conf 내용과 현재 ip 대역이 다르면 에러 -> 네트워크먼저

4. DHCP 서비스 재시작

systemctl restart isc-dhcp-server 명령어를 사용하여 DHCP 서비스를 재시작

```
root@sheep:~# systemctl restart isc-dhcp-server
root@sheep:~#
```

이후 systemctl status isc-dhcp-server 명령어를 통해 DHCP 서버가 active(running)

상태임을 확인

```
root@sheep:~# systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-09-09 09:33:38 UTC; 41s ago
     Docs: man:dhcpd(8)
    Main PID: 4211 (dhcpd)
      Tasks: 1 (limit: 4548)
    Memory: 3.7M (peak: 3.8M)
       CPU: 10ms
    CGroup: /system.slice/isc-dhcp-server.service
            └─4211 dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcpd.conf

Sep 09 09:33:38 sheep sh[4211]: Wrote 0 leases to leases file.
Sep 09 09:33:38 sheep dhcpd[4211]: PID file: /run/dhcp-server/dhcpd.pid
Sep 09 09:33:38 sheep dhcpd[4211]: Wrote 0 leases to leases file.
Sep 09 09:33:38 sheep dhcpd[4211]: Listening on LPF/ens33/00:0c:29:fd:d9:24/172.16.23.0/24
Sep 09 09:33:38 sheep sh[4211]: Listening on LPF/ens33/00:0c:29:fd:d9:24/172.16.23.0/24
Sep 09 09:33:38 sheep sh[4211]: Sending on LPF/ens33/00:0c:29:fd:d9:24/172.16.23.0/24
Sep 09 09:33:38 sheep sh[4211]: Sending on Socket/fallback/fallback-net
Sep 09 09:33:38 sheep dhcpd[4211]: Sending on LPF/ens33/00:0c:29:fd:d9:24/172.16.23.0/24
Sep 09 09:33:38 sheep dhcpd[4211]: Sending on Socket/fallback/fallback-net
Sep 09 09:33:38 sheep dhcpd[4211]: Server starting service.
(lines 1-21/21 (END))
```

5. VPC 에서 할당받은 IP 확인

ip dhcp 명령어를 통해 vpc 에서 할당받은 ip 확인

```
PC1>
PC1> ip dhcp
DORA IP 172.16.23.20/24 GW 172.16.23.30

PC1> sh ip

NAME          : PC1[1]
IP/MASK        : 172.16.23.20/24
GATEWAY        : 172.16.23.30
DNS            : 8.8.8.8
DHCP SERVER    : 172.16.23.35
DHCP LEASE     : 596, 600/300/525
DOMAIN NAME    : example.org
MAC            : 00:50:79:66:68:00
LPORT          : 10002
RHOST:PORT     : 127.0.0.1:10003
MTU:           : 1500

PC1> █
```